

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 20

Code of Conduct (192425119)

I N.R. Yasaswini (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ARM

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing

Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

CONCEPT MAP-1

CLOUD COMPUTING

FUNDAMENTALS

Central Concept - Cloud Computing

Definition -

Cloud computing is the delivery of computing resources such as servers, storage, databases, networking, software & analytics over the internet. It allows users to access services whenever needed without owning physical infrastructure.

Characteristics -

- * On-demand self-service.
- * Broad network access.
- * Resource pooling.
- * Rapid elasticity & scalability.
- * Measured service (pay-as-you-go model)

Benefits -

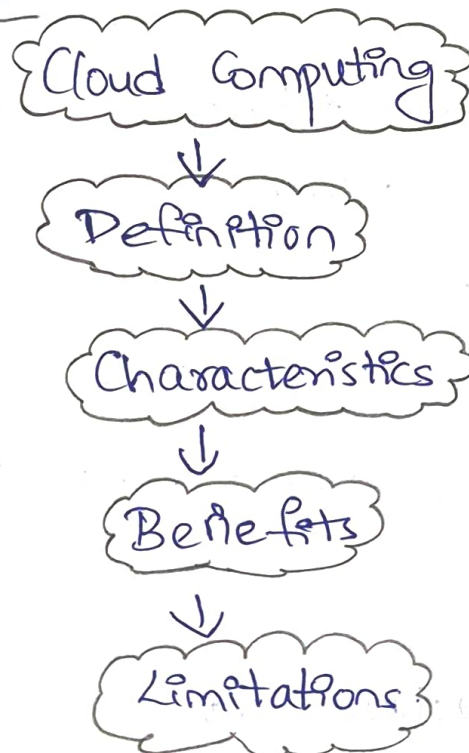
- ★ Lower infrastructure cost

- ★ High availability & reliability.
- ★ Easy scalability.
- ★ Flexibility and remote access.
- ★ Automation software updates.

Limitations -

- Security and privacy concerns.
- Internet dependency.
- Vendor lock-in.
- Possible service downtime.

Map Flow -



CONCEPT MAP-2

EVOLUTION OF CLOUD COMPUTING

Central Concept.- Evolution of Cloud Computing.

Hardware Evolution

(Powerful processors, storage, and networking)



Internet Evolution

(Faster communication and global connectivity)



Distributed Systems

(Multiple computers work together & share resources)



Virtualization

(Hypervisors create multiple virtual machines on one physical server).



Cloud Computing

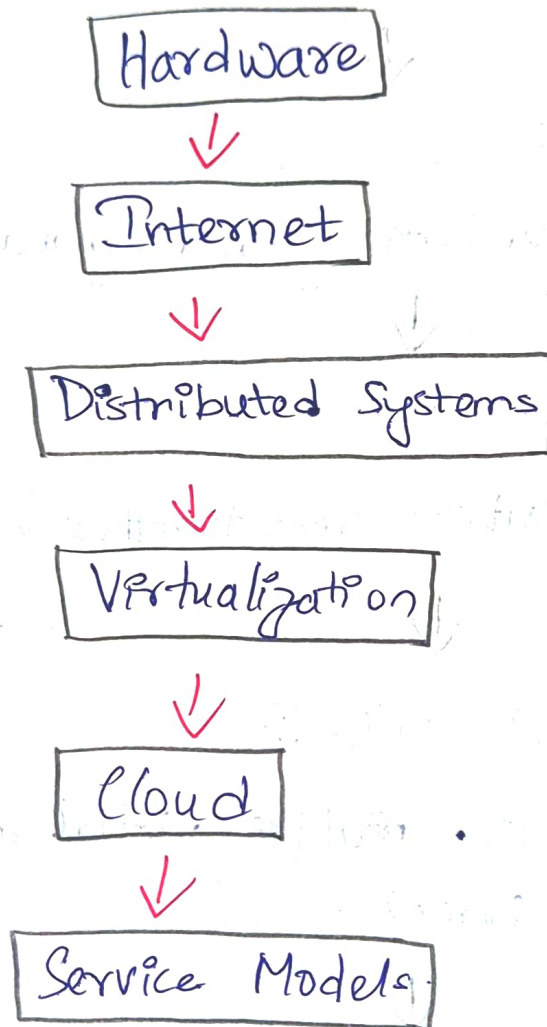
(Shared resources are delivered over the internet)



Service Delivery

(IaaS, PaaS, SaaS provide computing services to users).

Hierarchy :



CONCEPT MAP-3

SERVICE MODELS

Central Concept - Cloud Service Models

IaaS (Infrastructure as a Service)

(Provide manages servers, storage, and networking. User manages OS, applications, and data).



PaaS (Platform as a Service)

(Provider manages infrastructure, OS, runtime, and development tools. User develops and deploys applications).



SaaS (Software as a Service)

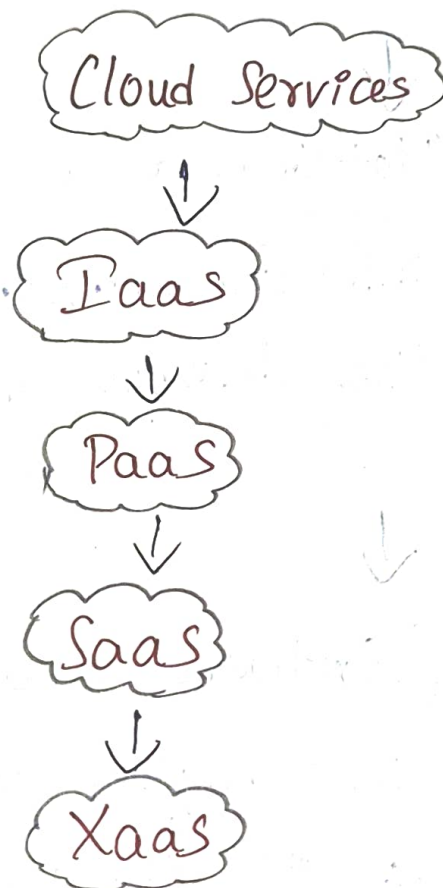
(Provider manages the complete software applications. Users simply access it through a browser or mobile app).



XaaS (Anything as a Service)

(Represents all cloud-based services such as Databases as a Service, Security as a Service, Storage as a Service, and many more).

Hierarchy -

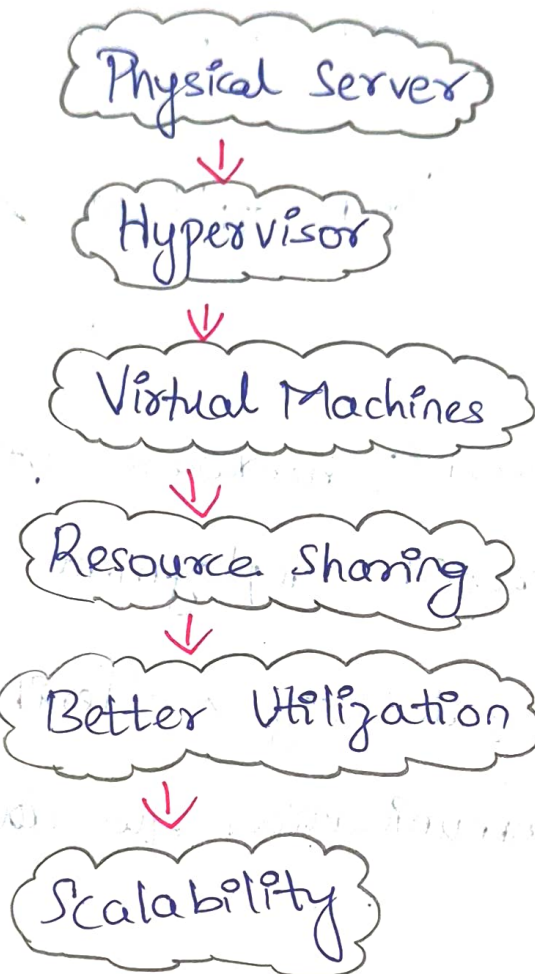


CONCEPT MAP-4

VIRTUALIZATION and WEB SERVICE AS ENABLERS of CLOUD

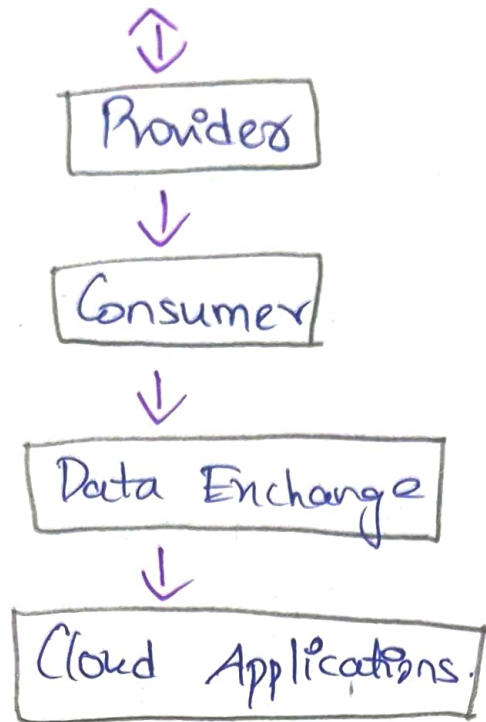
Central Concept - Virtualization & Web services.

Virtualization -



Web Services -





Importance -

- * Enables efficient hardware utilization.
- * Supports rapid deployment of applications.
- * Improves scalability & flexibility.
- * Allows communication b/w cloud services using API.

Hierarchy -

Virtualization + Webservices →

Cloud Deployment → Scalable Cloud
Environment.